higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

COMPUTER HARDWARE AND SOFTWARE
NQF LEVEL 3

(10041013)

26 February 2019 (X-Paper)
09:00–12:00

This question paper consists of 9 pages.

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TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use black or blue ink only.
5. Rule off across the page after each completed question.
6. Start each section on a NEW page.
7. Write neatly and legibly.

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SECTION A

QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Load testing is used to test the IT system functions under expected operating conditions.

1.2 A PC cannot fully function if all peripherals are not connected.

1.3 Fluctuations in power cause an unsuccessful computer installation.

1.4 The RAM stores data temporarily.

1.5 A main frame computer processes huge volumes of data.

1.6 Firmware is the programme installed in a computer.

1.7 A UPS is the main cause of power fluctuations in a computer.

1.8 A technical manual contains a section for troubleshooting.

1.9 A computer problem is defined by the identification of its parameters.

1.10 Diagnostic programmes are different from diagnostic software.

(10 × 1) **[10]**

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QUESTION 2

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–M) next to the question number (2.1–2.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
2.1	The unit that performs all the calculations in the computer	A	modem
2.2	A computer which is accessed by one person at a time	B	stress testing
2.3	A device that connects computers with a telephone line	C	ALU
2.4	A small high performance memory between CPU and RAM	D	expert system
2.5	A process that is used to perform system checks on the installation environment	E	third party
2.6	A test that is done to test the system response at unusually high or peak loads	F	installation wizard
2.7	One of the software resources required during software installation	G	internet
2.8	A signed document which binds the manufacturer should anything go wrong with the computers	H	user's support agreement
2.9	The term used for the user when the computer problem has moved from the technician to the technical support	I	stand-alone
2.10	A system where expert help about computer problems is readily available	J	Cache
		K	system testing
		L	UPS
		M	SRAM

(10 × 1) **[10]**

QUESTION 3

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (3.1–3.10) in the ANSWER BOOK.

3.1 A valuable information source which can be used to solve computer problems which are not easy to detect.

A Manufacturer's technical staff
B User's support agreement
C User manual
D Lease information

3.2 The … is the call centre of the computer industry.

A database
B server
C storage
D helpdesk

3.3 Problems which an average computer user cannot solve are … problems.

A environmental
B technical
C manual
D physical

3.4 A type of testing used to expose faults in the interaction between integrated units:

A Connectivity testing
B Load testing
C Stress testing
D Integration testing

3.5 Documentation which is important when planning the installation of any computer equipment.

A Estimation table
B Review procedures chart
C Lease agreement
D User specification requirements

3.6 The user can choose from … application software during installation.

A primary
B secondary
C standard
D pirated

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3.7 A kind of software that is embedded in a hardware device:

A Hardware
B Firmware
C Software
D Malware

3.8 The chip which stores computer settings such as date and time:

A CMOS
B RAM
C DOS
D EPROM

3.9 Peripherals such as a mouse and keyboard are found in the … of the computer.

A internal parts
B middle
C external parts
D memory

3.10 The static charge which is generated between two materials by friction:

A Triboelectric
B Conventional current flow
C Frictional force
D Electron flow

(10 × 1) **[10]**

TOTAL SECTION A: 30

SECTION B

QUESTION 4

Software is one of the basic elements of computer architecture. It is a set of instructions and associated documentation that tells a computer what to do or how to perform a task.

Explain FIVE different types of software. (5 × 2) **[10]**

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QUESTION 5

5.1 State SIX factors that will affect the performance of a client-server network. (6)

5.2 Define the following terms:

5.2.1 Benchmarking

5.2.2 User-acceptance sign-off

(2 × 2) (4)

5.3 Explain the functions of the following motherboard ports:

5.3.1 Parallel ports

5.3.2 Serial ports

5.3.3 USB ports

(3 × 2) (6)

5.4 What is a *hub*? (2)

5.5 Explain how a hub functions. (2)

[20]

QUESTION 6

State any FOUR minimum system requirements for installing Windows 7 or higher **[4]**

QUESTION 7

You are employed as a computer technician at a TVET college. The college has several computer laboratories for students as well as several PCs for staff. Your duties are to install system software and application software as well as repairing computer systems.

7.1 Discuss what you should do in order to plan the installation of system software and application software. (10)

7.2 Explain THREE functions of a device driver. (6)

7.3 Discuss the functions of the central processing unit (CPU). (4)

[20]

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QUESTION 8

8.1 Planning for the installation of a computer system is a very critical process because it determines the time and resources required for an installation.

8.1.1 Explain FIVE aspects that can help you assess whether the specifications for installing a new computer system are feasible or not. (5 × 2) (10)

8.1.2 State TWO advantages of burn-in testing. (2)

8.1.3 Explain the term *sanity testing* (2)

8.2 State THREE procedures which are to be considered when receiving notifications of problems from computer users. (3)

8.3 State any FOUR problem-solving strategies which are used when solving computer problems. (4)
[21]

QUESTION 9

9.1 Give FIVE reasons which may cause installation of computer peripherals to fail. (5)

9.2 When testing an IT system, a testing plan should be prepared according to the system requirements specifications that the system is expected to meet but not limited to.

List FIVE items that are included in the plan. (5)
[10]

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QUESTION 10

10.1 List any FIVE preventive maintenance measures which should be implemented to reduce computer maintenance problems. (5)

10.2 State SIX causes of environmental computer problems. (6)

10.3 How is troubleshooting different from a diagnosis? (4)

10.4 Which information sources can you use to identify problems that other computer users experienced before? (5)
[20]

QUESTION 11

11.1 List FIVE features of a proper dispatch procedure of a computer. (5)

11.2 State THREE sources which can provide expert help. (3)

11.3 State what can be done if a wireless mouse is not operating effectively. (3)

11.4 Mandy's computer takes long to open applications or to respond to user actions.

State FOUR possible solutions to this problem. (4)
[15]

TOTAL SECTION B: 120
GRAND TOTAL: 150

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